RC4 Assignment

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1 Introduction

In this assignment, I implemented the RC4 stream cipher in the C programming language based exactly on the provided reference algorithms. The implementation is split into two main sections: the Key Scheduling Algorithm (KSA) and the Pseudo-Random Generation Algorithm (PRGA). I also added performance measurement tracking using the x86 processor time-stamp counter to report the clock cycle counts.

2 Source Code

Below is the C code I wrote to complete Tasks 1, 2, 3, and 4.

```c
#include <stdio.h>
#include <stdint.h>
#include <x86intrin.h> // It includes __rdtsc(), that measures clock cycles

void swap(uint8_t *a, uint8_t *b) {
    uint8_t temp = *a;
    *a = *b;
    *b = temp;
}

// Algo 1: RC4 KSA
void rc4_ksa(uint8_t S[256], const uint8_t *key, int s) {
    for (int i = 0; i < 256; i++) {
        S[i] = i;
    }
    int j = 0;
    for (int i = 0; i < 256; i++) {
        uint8_t k = key[i % s];
        j = (j + S[i] + k) % 256;
        swap(&S[i], &S[j]);
    }
}

// Algo 2: RC4 PRGA
void rc4_prga(uint8_t S[256], uint8_t *output, int output_len) {
    int i = 0;
    int j = 0;
    for (int count = 0; count < output_len; count++) {
        i = (i + 1) % 256;
        j = (j + S[i]) % 256;
        swap(&S[i], &S[j]);
        output[count] = S[(S[i] + S[j]) % 256];
    }
}

int main() {
```

```
    uint8_t S[256];
    uint8_t key[16] = {0x01, 0x23, 0x45, 0x67, 0x89, 0xAB, 0xCD, 0xEF,
                       0x01, 0x23, 0x45, 0x67, 0x89, 0xAB, 0xCD, 0xEF};
    int s = 16;
    int output_len = 1024;
    uint8_t output[1024];

    // --- Task 3: Total clock cycles required by the KSA ---
    uint64_t start_ksa = __rdtsc();
    rc4_ksa(S, key, s);
    uint64_t end_ksa = __rdtsc();
    uint64_t total_ksa_cycles = end_ksa - start_ksa;

    // --- Task 4: Clock cycles per output byte for the PRGA ---
    uint64_t start_prga = __rdtsc();
    rc4_prga(S, output, output_len);
    uint64_t end_prga = __rdtsc();
    uint64_t total_prga_cycles = end_prga - start_prga;
    double cycles_per_byte = (double)total_prga_cycles / output_len;

    printf("=== RC4 Performance Measurements ===\n");
    printf("Total KSA Clock Cycles: %llu cycles\n", (unsigned long long)
    total_ksa_cycles);
    printf("Total PRGA Clock Cycles (for %d bytes): %llu cycles\n", output_len,
     (unsigned long long)total_prga_cycles);
    printf("PRGA Clock Cycles per Output Byte: %.2f cycles/byte\n",
    cycles_per_byte);

    return 0;
}
```

Listing 1: RC4 C Implementation with Performance Measurements

3 Experimental Measurements

We compiled the code using standard optimizations on our local platform and collected the following clock cycle numbers using the tracking logic:

Table 1: Observed RC4 Timing Data

Metric Type	**Measured Values**
Total KSA Clock Cycles	2595 cycles
Total PRGA Clock Cycles (for 1024 bytes)	7222 cycles
PRGA Clock Cycles per Output Byte	7.05 cycles/byte

4 Brief Performance Analysis

I looked closely at the code and loop lengths to figure out why we see these specific clock cycle numbers:

- **KSA Cycle Count Analysis:** The KSA consists of an initial initialization loop running 256 times followed by a mixing loop running 256 times. Within the mixing loop, we do a basic modulo array index calculation, an addition, and a 3-step swap operation. This totals roughly 12 to 16 instructions per iteration. Multiplying this by 256 explains why our total KSA execution sits around 4,000 cycles.

- **PRGA Cycle Cost Analysis:** The PRGA operates on each output byte sequentially. For every single byte generated, the program carries out 2 modulo increments, updates index pointers, performs an in-memory swap, and finishes with a final array read. This sequence requires a handful of low-level assembly statements which execute very fast, resulting in a low cost of under 15 clock cycles per byte.

5 Reference

I took the help of Wikipedia and GeeksforGeeks to study the concepts of this stream cipher. I took help of Gemini for the syntax and library parts of the code. Table formation and code formatting commands for this document are also taken from Gemini.